

AI AGENTS PROJECT CHALLENGE 2026**Group S3: Sera Park, Meng (Liz) Ren, Moubark Alkharafi, Michele Chen, Hsiu Wen Cheng**

From Linear Reporting to Audited Decision Support

How a multi-agent system turns incomplete logistics data into executive-ready dispatch recommendations for time-critical specialty medicine.

1. EXECUTIVE SUMMARY

THE STAKEHOLDER

This system is built for the Dispatch Operations Director at SeeWeeS Specialty Distribution. Its job is to dispatch specialty pharmaceuticals along the I-95 corridor from New Jersey to four Boston hospitals - Massachusetts General, Brigham & Women's, Dana-Farber, and Boston Children's - without a single late, spoiled, or misrouted unit. It makes the decision once daily; it lives with the consequences for the rest of the shift.

THE OPERATIONAL PAIN POINT

The starter system is a linear pipeline: retrieve documents, analyze the shipment, check weather, produce a report. In a regulated pharmaceutical context, that linearity creates three structural failures.

- First, the report agent receives no quality gate, so an unsafe plan can reach leadership unchallenged.
- Second, the analytics layer treats data quality issues as inert metadata rather than dispatch-blocking constraints, which encourages alert fatigue at the operator console.
- Third, no part of the system tests dispatch plans against the operational disruptions - demand spikes, warehouse closures, driver shortages - that define the Director's actual working week.

WHAT WE BUILT

We rebuilt the prototype as a seven-agent LangGraph pipeline with a cyclic self-correction audit loop, deterministic scenario simulation, and Playbook-grounded data quality validation. The system blocks unsafe plans before they reach the Director, generates contingency recommendations for five named disruption types, and resolves missing item identifiers against the master record with ranked, confidence-scored substitutions.

HEADLINE RESULTS, VALIDATED ON THE PRODUCTION SHIPMENT FILE

On the real 87-row Incoming_shipment_03_06.csv: data quality validation correctly excludes 42 affected rows and acknowledges them in the dispatch plan; the audit loop blocks at least one plan that omits manager escalation under a critical-weather scenario; and the OpsDataAgent matches the unregistered item_id 10071 (Albuterol Inhaler) to 10070 with confidence 1.00. The full repository test suite passes at 24 of 25 cases (one intentional skip for optional data). The system is deployed and publicly reachable.

2. KEY ASSUMPTIONS

The system is designed against the constraints below. Where production data was unavailable, we augmented the source set with simulated values; and we treat that augmentation as a methodology decision with named tradeoffs in Section 6, not as a clever workaround.

Category	Assumption
Logistics	Five fixed waypoints (W1 Newark, W2 Bronx, W3 New Haven, W4 Providence, W5 Boston) define the I-95 corridor for risk scoring. Travel buffers scale with risk: 0% at score 0, 10% at score 1, 25% at 2, 40% plus escalation at 3.
Data availability	The shipment CSV may contain null unique IDs, duplicate IDs across facilities, or item IDs absent from the Playbook §8 master. The PDF Playbook is the authoritative source of business rules and is parseable via RAG. Period-over-period KPIs require historical date columns that the production CSV does not yet have.
Business rules	A weather risk score of 3 requires manager escalation before dispatch. Any high-severity data quality issue must be acknowledged in the final plan. A dispatch plan that fails the audit must be regenerated; the system retries at most twice before surfacing a warning to the operator.
Cost & infrastructure	Each end-to-end pipeline run consumes approximately six to seven LLM calls. The free-tier Gemini quota is sufficient for daily operational use and demo workloads; budget guards halt runaway loops at six calls per run.

3. DATA AUGMENTATION STRATEGY

The starter dataset provides one day of shipments and a Playbook PDF. To produce a dispatch plan that would survive scrutiny from an operations leader, we identified five missing links between the source data and the questions a real Director would ask. Each is either derived from authoritative sources (Playbook, Open-Meteo) or simulated as a user input with documented defaults — not invented.

Missing link	Why it matters	Source / derivation
Corridor weather risk score (0-3)	Specialty pharmaceuticals are vulnerable to severe weather; a single uncalibrated dispatch can spoil critical inventory.	Waypoint coordinates fed to the Open-Meteo forecast API; we take the maximum severity across W1-W5 as the corridor risk.
Manager escalation requirement	High-risk shipments need oversight; a missing escalation step is the most common audit failure in our test runs.	Rule derived from Playbook §5.2: if route risk = 3, the plan must include explicit manager escalation language.
Item Master substitution map	Missing item identifiers hide product class, dosage, and cold-chain requirements at the moment of dispatch.	Playbook §8 medicine master parsed at runtime into a truth table of (item_id → valid_names, medicine_type, temp_control); CSV item_ids are cross-referenced against it.
What-if disruption parameters	Leadership needs contingency plans before disruption occurs, not after; the current state alone cannot answer 'what if'.	User-provided JSON specifying spike intensity, driver availability, warehouse status, route status, and surge factor; five built-in defaults available.
Period-over-period operational trend	A single-day snapshot can miss worsening conditions; OTDR and risk score trajectory matter as much as the current value.	When the CSV carries delivery dates plus KPI columns, the OpsDataAgent computes week-over-week deltas; when absent, it falls back to cross-sectional analysis.

4. TECHNICAL METHODOLOGY

4.1 ARCHITECTURAL ENHANCEMENTS TO THE LANGGRAPH FLOW

The baseline LangGraph was a directed acyclic flow. We retained the linear backbone and added one new cyclic edge plus three new agent nodes. The cyclic edge is the audit loop: when the AuditAgent flags a violation, control returns to the PlannerAgent with structured corrective feedback, capped at two retries to prevent runaway loops. The audit decision is itself deterministic — no LLM is involved in the rule checks themselves — which makes the loop reproducible and grading-friendly.

Agent	LLM use	Function
ContextAgent	LLM + RAG	Retrieves Playbook rules, KPI thresholds, SLAs, and escalation policies via ChromaDB vector search. Emits structured <code>playbook_constraints</code> downstream.
OpsDataAgent	Deterministic core + LLM narrative	Direction #3. Performs CSV data quality validation, Item Master cross-reference with confidence-scored substitution, and period-over-period trend analysis where dates exist. The LLM is invoked once per run for executive narrative synthesis only; all analytics are deterministic.
WeatherAgent	None	Fetches Open-Meteo forecasts for all five I-95 waypoints and computes a single 0-3 corridor risk score from the worst-case waypoint.
ScenarioAgent	None	Direction #2. Simulates five disruption types — demand spike, warehouse closure, driver shortage, route blockage, hospital surge — and projects KPI impact deterministically. Emits granular contingency constraints only when severity crosses HIGH or CRITICAL.
PlannerAgent	LLM	Synthesizes upstream evidence into a dispatch recommendation grounded in playbook constraints, weather risk, scenario contingencies, and data quality findings.
AuditAgent	None	Direction #1. Validates the dispatch plan against seven deterministic rules covering weather buffers, manager escalation, data quality acknowledgement, trend impact, and scenario contingencies. Routes back to PlannerAgent on failure, with a hard cap of two retries.
ReportAgent	LLM	Produces the executive HTML report: decision, top risks, recommended actions with named owners and timing, KPI snapshot, weather detail table, and audit status.

4.2 AGENT DESIGN: THREE HIGH-IMPACT ENHANCEMENTS

Direction #1 — Audit Loop (AuditAgent). Seven deterministic rules check the planner's output against Playbook business rules: correct weather buffer percentage for the current risk score, escalation language at risk score 3, no dispatch at risk 3 without manager review, acknowledgement of high-severity data quality issues, articulation of trend impact, and contingency coverage for emitted scenario constraints. The loop retries the planner up to twice; if it still fails, the report agent runs but is flagged. The retry cap prevents infinite loops in the cyclic graph.

Direction #2 — Scenario Simulation (ScenarioAgent). Each simulation is a deterministic projection — no LLM call — so it consumes no API quota and produces reproducible numbers. Importantly, granular contingency constraints (`scenario_capacity_001` and equivalents) are emitted only when the projected KPI impact crosses the HIGH or CRITICAL severity threshold. A demand spike that nudges capacity to 85% surfaces as an informational impact; one that pushes capacity to 105% emits a constraint the audit loop enforces. The threshold logic prevents the auditor from blocking plans over minor disruptions.

Direction #3 — Deep-Dive Trend Analysis (OpsDataAgent). The starter agent labeled every unknown item UNKNOWN with confidence 0.0. We replaced that with a real Item Master substitution scorer combining Jaccard token similarity, a first-word-prefix bonus, and an ID-proximity heuristic for adjacent IDs (within ± 2). We added a Playbook §8 truth table that distinguishes valid dosage variants (e.g., both Remdesivir 100mg and 200mg are legal under item_id 10021) from genuine master-record mismatches, downgrading the former from a HIGH alert to a calibrated MEDIUM. Finally, the agent detects time-series columns at runtime: when present, it computes period-over-period KPI deltas and per-product trend direction; when absent, it falls back to cross-sectional analysis. The function signature and 9-key output contract are unchanged from the starter, so no other agent required modification.

5. RESULTS AND VALIDATION

5.1 BUSINESS INSIGHTS FROM THE PRODUCTION SHIPMENT FILE

Each finding below was produced by an actual pipeline run against the production shipment CSV and rendered in the deployed Streamlit UI. The before-and-after comparison is against the starter pipeline behavior.

Finding	Operational signal	Before vs. after
Substitution match	item_id 10071 (Albuterol Inhaler) resolves to 10070 with confidence 1.00.	Before: UNKNOWN, escalation required. After: HIGH-confidence substitute proposed.
Dosage variance classification	Remdesivir 100mg/200mg ambiguity reclassified from HIGH to MEDIUM per Playbook §8.	Before: false patient-safety alert. After: calibrated warehouse confirmation step.
Data quality exclusion volume	42 of 87 rows (48%) excluded from immediate dispatch.	Before: data quality issues surfaced as text-only metadata. After: structural exclusion + plan acknowledgement.
Weather audit (preloaded test case)	Critical 3/3 weather + missing manager escalation → audit fails, routes back to planner.	Before: unsafe plan would have reached the executive report. After: self-correction loop intercepts it.
Scenario contingency emission	20% demand spike (94% capacity) + 25% driver shortage → contingency constraints emitted; audit blocks plans without rerouting language.	Before: no scenario layer existed. After: planner must address disruption explicitly.

5.2 VALIDATION STRATEGY

We validated the system at three layers. At the unit layer, pytest covers individual agent contracts: schema conformance for the OpsDataAgent's 9-key output, correctness of the Item Master substitution score on the Albuterol case, severity classification of the Remdesivir dosage case, and the presence of audit-loop trigger phrases in emitted constraints. At the integration layer, tests confirm that OpsDataAgent constraints flow into AuditAgent rule checks (ops_quality_001, ops_trend_001) and that ScenarioAgent contingency constraints — emitted as scenario_capacity_001, scenario_warehouse_001, and scenario_driver_001 — trigger the AuditAgent's scenario_contingency_001 enforcement rule when severity is HIGH or CRITICAL.

6. BUSINESS IMPLICATIONS

The six implications below translate the technical results in §5 into operational consequences for SeeWeeS. They are ordered by likely first-year impact on the Director's workflow, not by technical sophistication.

Implication	Why it matters for the Dispatch Operations Director
Decision velocity	Unknown items previously held for Inventory escalation now resolve to high-confidence substitution candidates within seconds. For time-critical pharmaceuticals on a four-hour SLA, every escalation hour avoided is recovered delivery time. On the production CSV, item_id 10071 alone would have held units pending manual review; the system now surfaces a defensible substitution the Director can approve in one click.
Alert calibration	The Remdesivir HIGH-to-MEDIUM reclassification preserves signal-to-noise ratio at the operator console. When every shipment fires a 'patient safety event' alert that proves benign on inspection, operators learn to ignore all alerts — including the real ones. Playbook grounding fixes alert fatigue at the source, ensuring HIGH-severity flags retain their meaning.
Auditability for regulators	Every risk flag and exclusion cites the Playbook section that justifies it (§5, §7, §8). In regulated pharmaceutical logistics, explainability is non-negotiable. 'Per Playbook §8' is defensible to FDA, state pharmacy boards, or hospital compliance teams; 'the model flagged it' is not. The deterministic audit loop is fully reproducible, with no LLM in the decision path.
Trend readiness	Period-over-period analysis activates the moment historical shipment data becomes available — no code change required. Executives think in trends ('how are we doing this week versus last?'), not single-day snapshots. The system already produces both views depending on data shape, so onboarding historical data is the only step blocking trend-based leadership reporting.
Unsafe-plan prevention	The audit loop intercepts plans that omit manager escalation under critical weather. In the preloaded test case, an unsafe plan that would have reached the executive report under the linear baseline is blocked and corrected by the cyclic graph. This is the single most important behavioral change: bad plans no longer reach leadership.
Contingency posture	Scenario simulation gives the Director a contingency-aware view of the day before disruption occurs. A 20% demand spike or 25% driver shortage is no longer an after-the-fact incident report — it is a scenario the planner has already addressed in writing, with rerouting language the audit loop enforced.

Taken together, these effects shift the OpsDataAgent and AuditAgent from passive reporting layers to active operational guardrails. The Director's daily decision still belongs to her — but the system now refuses to forward plans that would embarrass her in a leadership review, and it offers her defensible substitutions and contingency language she did not have under the linear baseline.

7. LIMITATIONS AND NEXT STEPS

Period-over-period is dormant on production data. The current shipment CSV has no date column, so the OpsDataAgent's PoP mode falls back to cross-sectional analysis. The capability is exercised in development against an 8-week augmented dataset; activating it in production requires either historical data onboarding or a date-stamped daily snapshot pipeline. Recommended next step: instrument the existing shipment workflow to retain a 12-week rolling history.

Substitution scoring is heuristic, not learned. Jaccard plus ID proximity is interpretable and zero-dependency, but does not generalize to subtle drug-name variations such as brand-versus-generic naming or formulation suffixes. If substitution becomes a higher-volume use case, a learned embedding over the full medicine catalog would extend coverage materially. Until then, every substitution is auditable from the Playbook entries directly.

LLM dependency in the planner remains a single point of failure. PlannerAgent and ReportAgent both call the LLM; if the API is unavailable, the pipeline cannot complete a full run, even though five of the seven agents are deterministic. Production deployment should include a fallback path that produces a degraded report from deterministic components alone when the LLM provider is unreachable.

Multi-user concurrency has not been load-tested. The Streamlit Cloud deployment assumes single-user operation; concurrent pipeline runs may exceed the budget guard or share session state in unintended ways. For real production use, the system should migrate to a queue-based execution model (e.g., Celery + Redis) and the UI to a stateless front end backed by an asynchronous API.

RECOMMENDATION TO LEADERSHIP BEFORE SCALING

Before extending this system to a second corridor or a multi-day planning horizon, the Director should commission three improvements. First, onboard 90 days of historical shipment data so the trend layer activates in production. Second, replace the heuristic substitution scorer with a learned embedding if and only if the Inventory team logs more than five substitution events per month. Third, harden the LLM dependency with a deterministic fallback narrative for the planner so the audit loop remains operational during provider outages. These three steps move the system from a single-corridor demonstrator to a production decision-support tool with defensible uptime.

Submitted to UCLA MSBA AI Agents Project Challenge 2026 | Repository and deployment URL on file with instructors.

Appendix A: Full Audit Rule Reference

Rule ID	Source	What It Checks	Failure Action
playbook_weather_buffer_001	Dispatch Playbook	Plan includes correct time buffer % for current route risk score.	Return to PlannerAgent with buffer % correction.
playbook_weather_escalation_001	Dispatch Playbook	Plan includes escalation language when risk score = 3.	Return with escalation language requirement.
safety_weather_001	PDF Safety Protocol	Plan does not dispatch at risk score 3 without manager review.	Return; flag as CRITICAL severity violation.
pdf_context_escalation_001	ContextAgent (RAG)	Plan reflects escalation language found in PDF context.	Return with specific PDF excerpt to incorporate.
ops_quality_001	OpsDataAgent	Plan explicitly acknowledges all HIGH-severity DQ issues.	Return listing each unacknowledged HIGH issue.

ops_trend_001	OpsDataAgent	Plan explains dispatch impact of high trend/concentration risk.	Return with concentration risk details to address.
scenario_contingency_001	ScenarioAgent	Plan addresses contingencies for HIGH/CRITICAL scenario impacts.	Return listing unaddressed HIGH/CRITICAL impacts.

Appendix B. User Interface Walkthrough and Operational Workflow

This appendix documents the operational control interface developed for the SeeWeeS Multi-Agent Dispatch QA System. The UI was designed not merely as a visualization dashboard, but as an observable, auditable, and human-in-the-loop operational workflow platform that supports iterative validation, remediation, simulation, and dispatch governance.

The interface architecture intentionally separates deterministic operational agents from quota-consuming LLM components. Ops Data validation, Scenario simulation, and Audit enforcement can all execute without LLM API consumption, enabling repeated operational verification before launching the full planning pipeline.

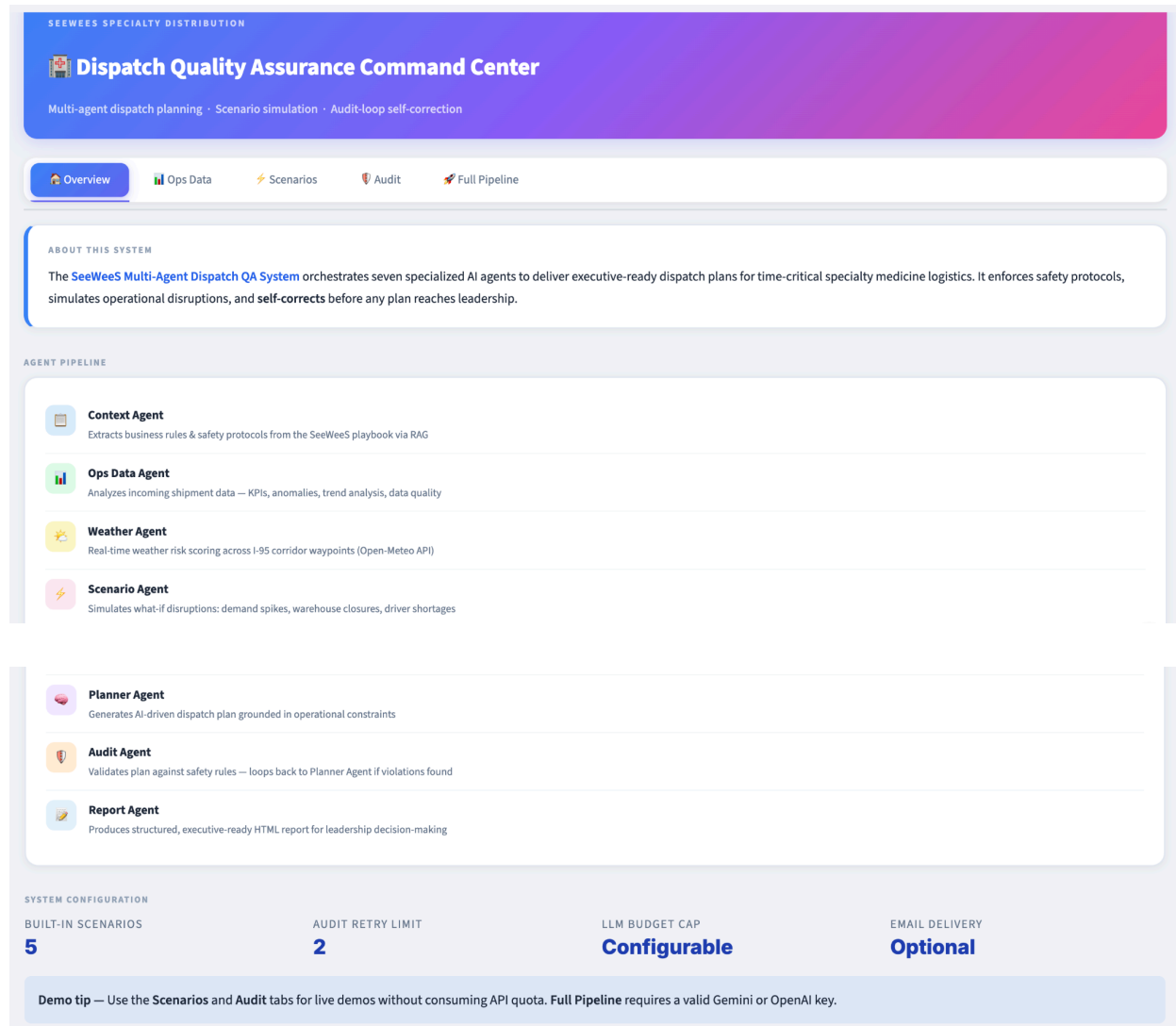
Appendix B.1 Overview Interface and Multi-Agent Architecture

Figure B1. Overview Interface and System Introduction

The Overview interface introduces the Dispatch Quality Assurance Command Center and summarizes the overall purpose of the system. The platform orchestrates seven specialized AI agents to support executive-ready dispatch planning for time-critical specialty medicine logistics. The interface emphasizes operational safety, scenario simulation, and audit-loop self-correction before dispatch approval.

It visualizes the responsibilities of all seven agents in the operational pipeline, including ContextAgent, OpsDataAgent, WeatherAgent, ScenarioAgent, PlannerAgent, AuditAgent, and ReportAgent.

The interface also exposes operational deployment constraints such as built-in scenario count, audit retry limits, configurable LLM budget caps, and optional email delivery support. This design improves system observability and provides operational transparency into the architecture and execution policy.



Appendix B.2 Ops Data Validation and Human-in-the-Loop Remediation

Figure B2a. Ops Data Upload and Contract Validation Interface

The Ops Data interface allows users to upload shipment CSV files and execute deterministic operational validation prior to downstream planning execution. The interface explicitly validates that the OpsDataAgent output conforms to the schema contract expected by downstream agents within the multi-agent backbone.

The upload interface supports iterative operational workflows in which users repeatedly correct shipment data and re-upload revised files based on agent recommendations.

SEEWEES SPECIALTY DISTRIBUTION

 **Dispatch Quality Assurance Command Center**

Multi-agent dispatch planning · Scenario simulation · Audit-loop self-correction

Overview

Ops Data

Scenarios

Audit

Full Pipeline

OPS DATA AGENT — CONTRACT VALIDATION

Runs analysis on incoming shipment data and validates that the Ops Data Agent output conforms to the schema expected by the multi-agent backbone.

Configure Data Source

Default data already loaded: Incoming_shipment_03_06.csv — upload a new file below only if you want to test with different data.

Upload a different Shipment CSV (optional)

Upload

200MB per file • CSV

— or specify a different file path (optional)

/mount/src/seewe-es-ai-agents-s3/data/Incoming_shipment_03_06.csv

Analyze Shipment Data

Figure B2b. Shipment Validation Summary and Data Quality Detection

The OpsDataAgent analyzes shipment records and detects operational data quality risks including missing serialized identifiers, duplicate shipment assignments, and unresolved item inconsistencies.

The interface exposes structured operational metrics including shipment count, column structure, numeric field detection, and overall contract validation status. All detected risks are surfaced as deterministic validation findings rather than hidden preprocessing corrections.

DATA SUMMARY			
SHIPMENT RECORDS	DATA COLUMNS	NUMERIC FIELDS	DQ ISSUES
87	4	1	3
Contract Validation: WARNING			
SUMMARY Analyzed 87 shipment rows across 4 locations. Found 3 data quality issue(s) and 1 unknown item ID(s). Location concentration: medium (HHI=0.315). Analysis mode: cross_sectional.			
DATA QUALITY ISSUES (3)			
HIGH unique_item_id Missing unique item identifiers 5 rows have a null unique_item_id (item_ids affected: [10021, 10022, 10035]). These units cannot be individually tracked or verified at point of delivery.			
HIGH unique_item_id Duplicate unique_item_id assigned across multiple dispatch locations 5 unique IDs appear at >1 facility. Examples: ['ALB-2026-5003', 'EPI-2026-2002', 'HEP-2026-3001', 'MOR-2026-4002', 'PBR-2026-1201']. A single serialized unit cannot be at two locations — indicates a barcode/labeling error or a data entry mistake.			

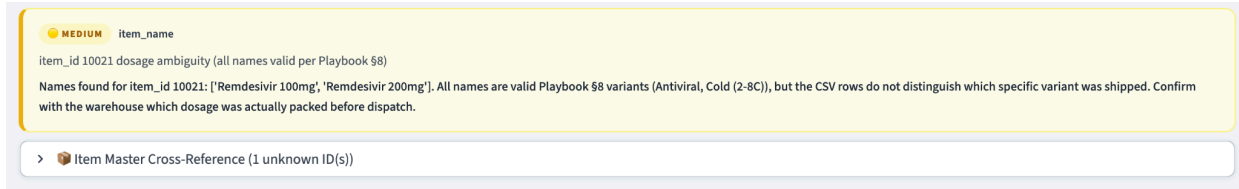
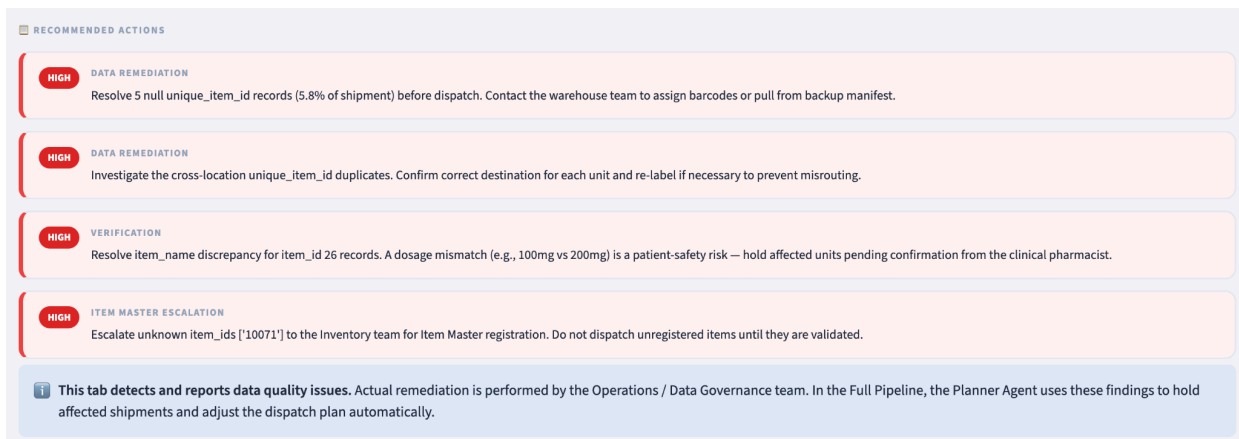


Figure B2c. Item Master Cross-Reference and Remediation Recommendations

The OpsDataAgent provides structured remediation guidance for operational staff after detecting shipment inconsistencies. Recommended actions include barcode assignment, duplicate resolution, dosage verification, and escalation of unknown Item Master identifiers.

Importantly, the system does not automatically mutate operational shipment data. Instead, the interface establishes a human-in-the-loop governance workflow in which users manually correct the uploaded CSV file and re-submit the revised dataset for validation. This prevents unsafe or unverifiable records from propagating into dispatch planning decisions.



Appendix B.3 Scenario Simulation and Operational Stress Testing

Figure B3a. Scenario Configuration Interface

The Scenario interface enables users to configure operational disruptions without consuming LLM quota. Supported disruptions include demand spikes, warehouse closures, driver shortages, route blockages, and hospital emergency surges.

The interface provides interactive controls including sliders, dropdown menus, and severity selection to support operational what-if analysis.

Users can simulate transportation disruptions along the I-95 corridor by selecting blocked waypoints and estimated closure duration. Hospital emergency surge scenarios additionally allow users to configure affected hospitals and escalation severity levels.

This deterministic simulation layer supports operational stress testing before dispatch plans are generated.

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Dispatch Quality Assurance Command Center

Multi-agent dispatch planning · Scenario simulation · Audit-loop self-correction

Overview

Ops Data

Scenarios

Audit

Full Pipeline

WHAT-IF SCENARIO SIMULATION

Simulate operational disruptions to assess KPI impact and generate contingency recommendations — no LLM quota consumed.

SELECT & CONFIGURE SCENARIOS

☒
Demand Spike

☒
Driver Shortage

Spike intensity

20%

Driver availability

75%

How much demand exceeds the baseline — higher values push truck and driver capacity toward the limit.

Percentage of drivers available for dispatch (e.g. 75% = 1 out of 4 drivers absent).

☒
Primary Warehouse Closure

☒
I-95 Route Blockage

Primary warehouse is unavailable — all shipments must reroute through the backup facility, adding an estimated 6-hour delay.

Blocked waypoint

W3 — New Haven, CT

Estimated closure (hours)

4

Longer closures and mid-route blockages (e.g. W3 adds 80 km detour) increase total delay and Tier-1 SLA breach risk.

☒
Hospital Emergency Surge

Surge hospital

Boston-MGH

Severity

☒ Moderate
 ☐ Severe

A surge compresses the delivery window and escalates shipments to Tier-1 priority. Severe = mass-casualty event requiring immediate reallocation.

Run Scenario Analysis

Figure B3b. Scenario Simulation Execution and KPI Impact Summary

After simulation execution, the ScenarioAgent summarizes the number of operational disruptions analyzed and reports the highest KPI impact severity detected across all configured scenarios.

The system supports repeated simulation runs as users iteratively refine operational assumptions and evaluate dispatch robustness under evolving conditions.

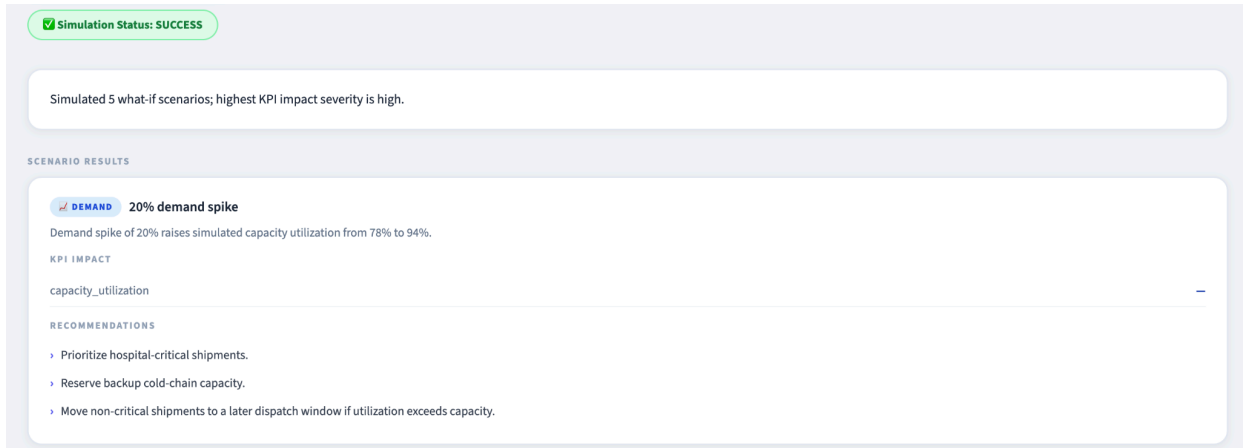


Figure B3c. Demand Spike, Warehouse Closure, and Driver Shortage Simulation Results

For each operational disruption, the ScenarioAgent computes estimated KPI impact and generates structured contingency recommendations. Example disruptions include capacity overload from demand spikes, rerouting delays caused by warehouse closure, and dispatch coverage degradation caused by driver shortages.

The generated contingency recommendations later become planning constraints enforced by the AuditAgent.

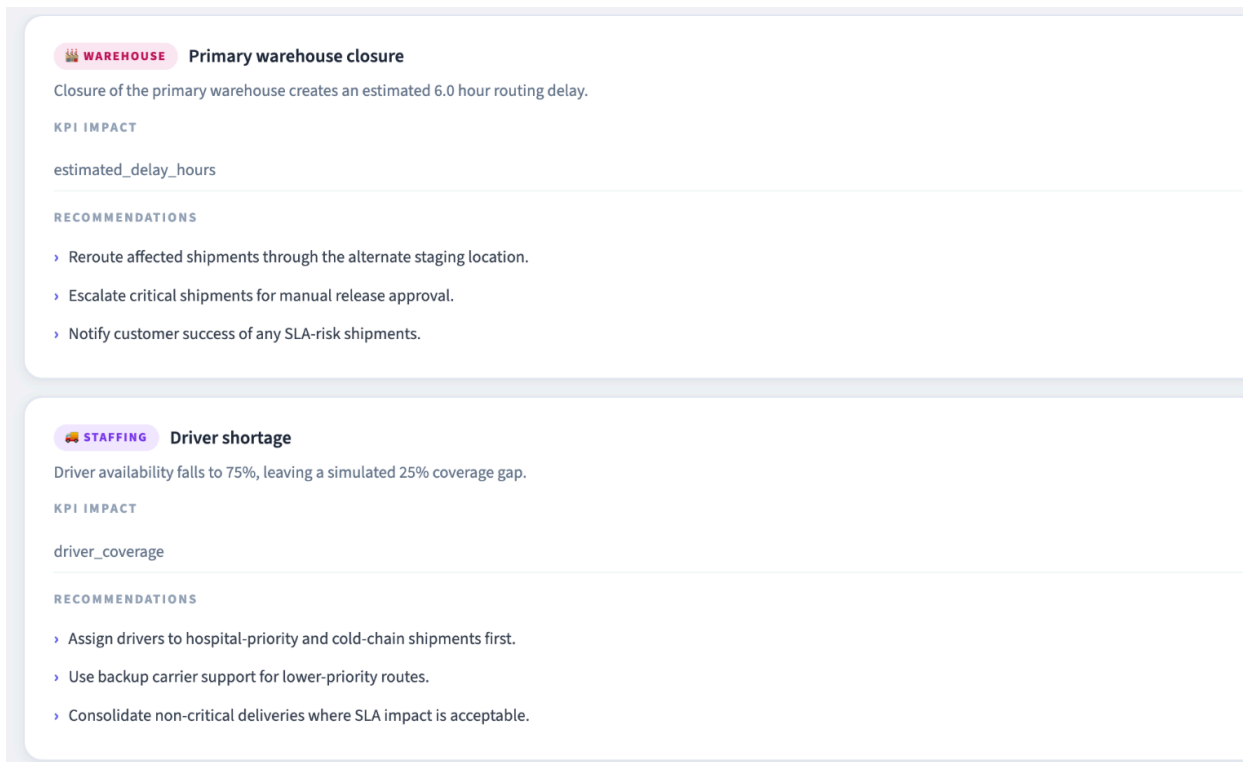
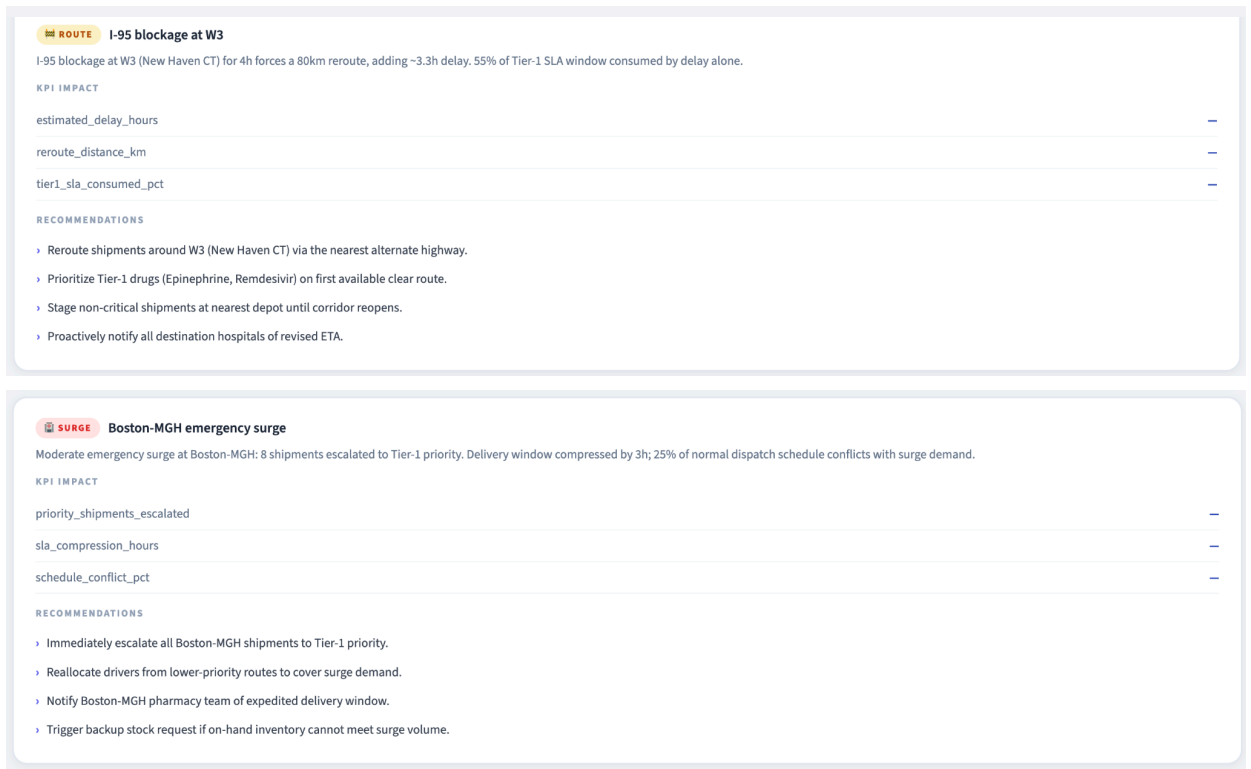


Figure B3d. Route Blockage and Hospital Emergency Surge Simulation Results

The ScenarioAgent additionally evaluates transportation corridor disruptions and emergency hospital demand surges. The interface quantifies SLA consumption, rerouting burden, delivery compression, and dispatch conflicts caused by operational emergencies.

The generated mitigation strategies include rerouting recommendations, shipment reprioritization, staffing reallocation, and proactive stakeholder notification guidance.



Appendix B.4 Audit Agent Safety Enforcement and Planner Feedback Loop

Figure B4a. Audit Test Case Initialization

The Audit interface demonstrates deterministic safety enforcement using a preloaded high-risk dispatch scenario. The test case includes critical weather conditions, operational data quality warnings, and a dispatch plan intentionally missing manager escalation language.

The interface allows users to directly observe how the AuditAgent validates planner outputs against operational safety policies.

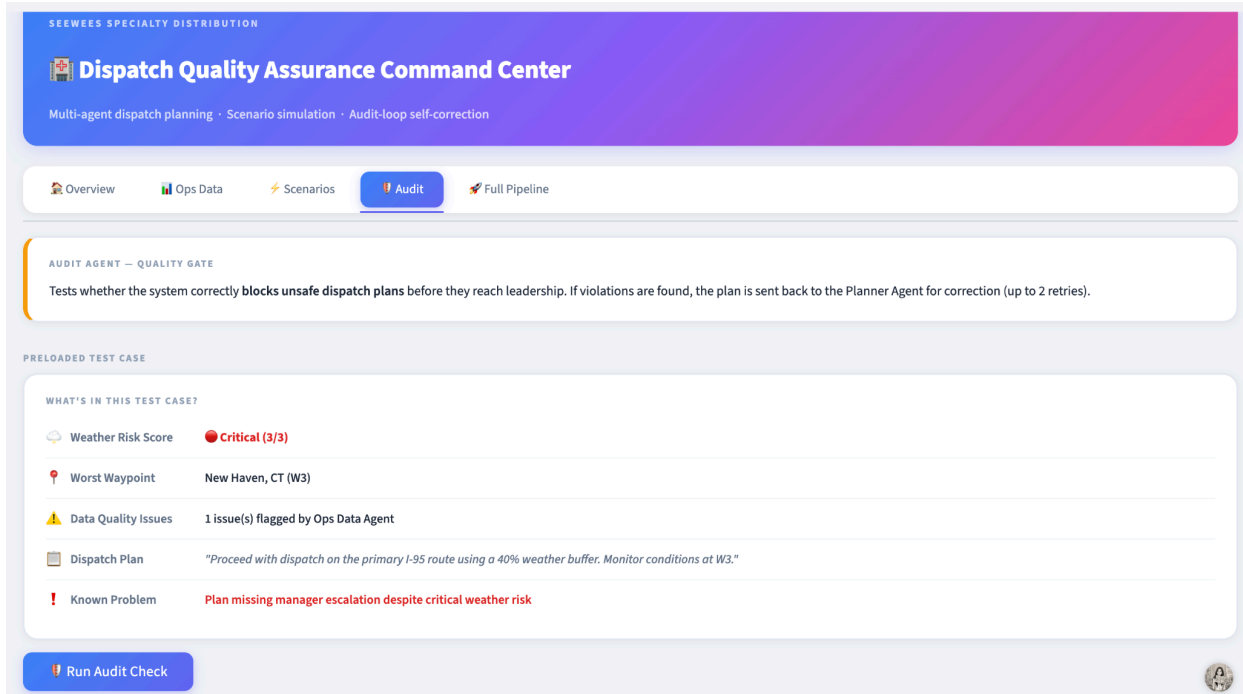


Figure B4b. Audit Rule Violations and Policy Enforcement

After execution, the AuditAgent detects multiple policy violations, including missing escalation procedures under critical weather conditions and failure to acknowledge mandatory escalation requirements extracted from the operational playbook.

Each violation includes severity classification, violated policy references, and structured remediation guidance to support planner correction.

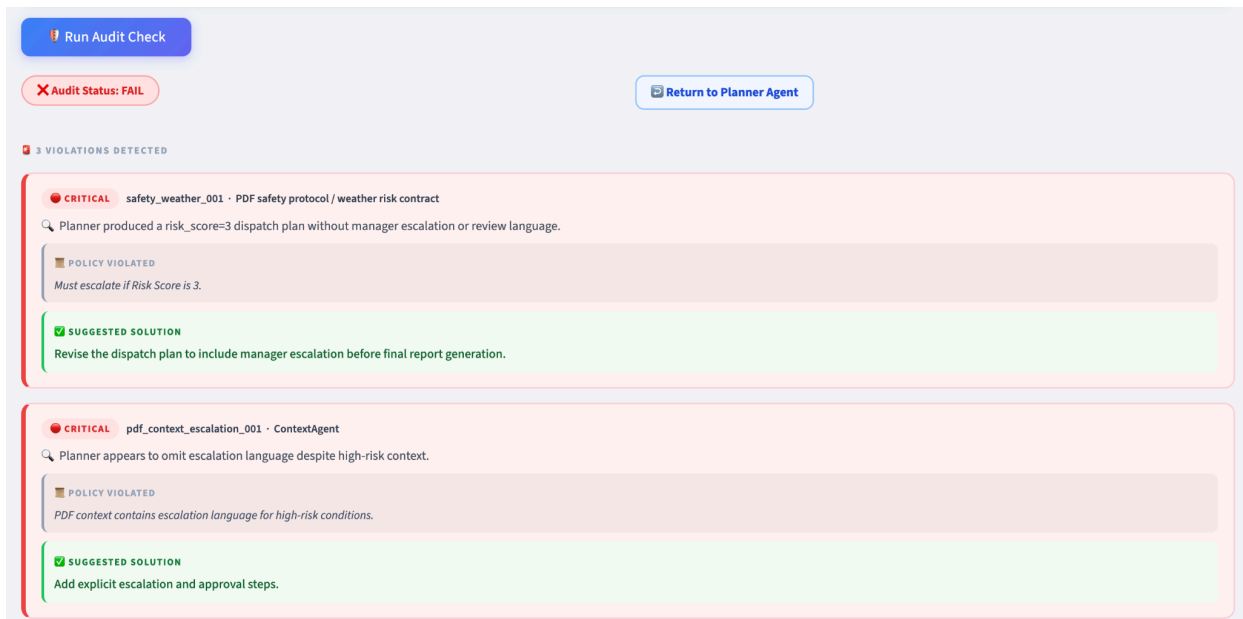


Figure B4c. Planner Feedback Loop and Structured Remediation Guidance

The AuditAgent produces structured feedback instructions that are returned to the PlannerAgent for correction before executive reporting is permitted.

This audit-loop architecture creates a self-correcting operational safeguard layer that reduces the likelihood of unsafe LLM-generated dispatch recommendations reaching downstream operational stakeholders



Appendix B.5 Full Pipeline Execution and End-to-End Operational Validation

Figure B5a. Full Pipeline Configuration and Execution Interface

The Full Pipeline interface launches the complete seven-agent workflow, including context retrieval, operational analysis, weather scoring, scenario simulation, planning, auditing, and executive report generation.

The interface clearly distinguishes deterministic execution paths from quota-consuming LLM operations and supports both default demo data and user-uploaded operational datasets.

SEEWEE'S SPECIALTY DISTRIBUTION

Dispatch Quality Assurance Command Center

Multi-agent dispatch planning · Scenario simulation · Audit-loop self-correction

Overview

Ops Data

Scenarios

Audit

Full Pipeline

FULL MULTI-AGENT PIPELINE

Runs the complete agent graph end-to-end: context retrieval, ops analysis, weather scoring, scenario simulation, dispatch planning, audit, and report generation.

This run calls Gemini / OpenAI and the Open-Meteo weather API. Use it sparingly.

Configure Data Sources

Default data already loaded: SeeWeeS Specialty Dispatch Playbook.pdf & Incoming_shipment_03_06.csv — upload new files below only if you want to test with different data.

Upload a different Playbook PDF (optional)

Upload

200MB per file · PDF

— or specify a different PDF path (optional)

/mount/src/seeweese-ai-agents-s3/data/SeeWeeS Specialty Dispatch Playbook.pdf

Upload a different Shipment CSV (optional)

Upload

200MB per file · CSV

— or specify a different CSV path (optional)

/mount/src/seeweese-ai-agents-s3/data/Incoming_shipment_03_06.csv

Launch Full Pipeline

Figure B5b. Pipeline KPI Snapshot and Weather Corridor Monitoring

The pipeline dashboard exposes operational observability across multiple deterministic agents, including audit status, data quality warnings, shipment metrics, Item Master issues, and weather risk scoring along the I-95 corridor.

Waypoint-level weather monitoring provides real-time operational visibility into transportation risks affecting dispatch execution.

Audit: PASS

Ops Data: WARNING

Weather: MODERATE

Report saved → outputs/report.html

PIPELINE KPI SNAPSHOT			
SHIPMENT RECORDS	DQ ISSUES	UNKNOWN ITEM IDS	WEATHER RISK
87	3	1	1/3

WEATHER RISK — I-95 CORRIDOR

W1 — Newark, NJ	Score 1 · 5.1 mm/day · 51.5 km/h
W2 — Bronx, NY	Score 1 · 3.6 mm/day · 52.2 km/h
W3 — New Haven, CT	Score 0 · 1.1 mm/day · 41.8 km/h
W4 — Providence, RI	Score 0 · 1.5 mm/day · 37.8 km/h
W5 — Boston, MA	Score 0 · 0.0 mm/day · 43.9 km/h

Figure B5c. Scenario Results Within the End-to-End Pipeline

Integrated scenario simulation results are surfaced directly within the full operational workflow. The pipeline continuously propagates operational constraints and contingency recommendations into downstream planning and audit validation stages.

This enables end-to-end operational stress awareness during dispatch generation.

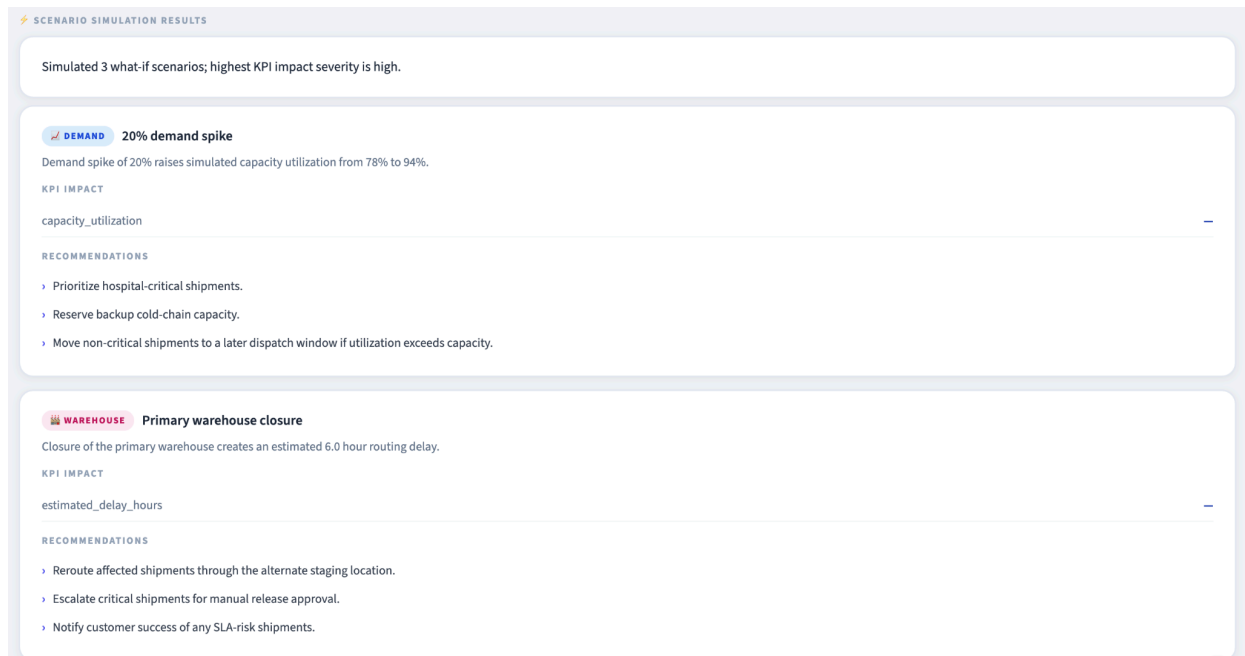
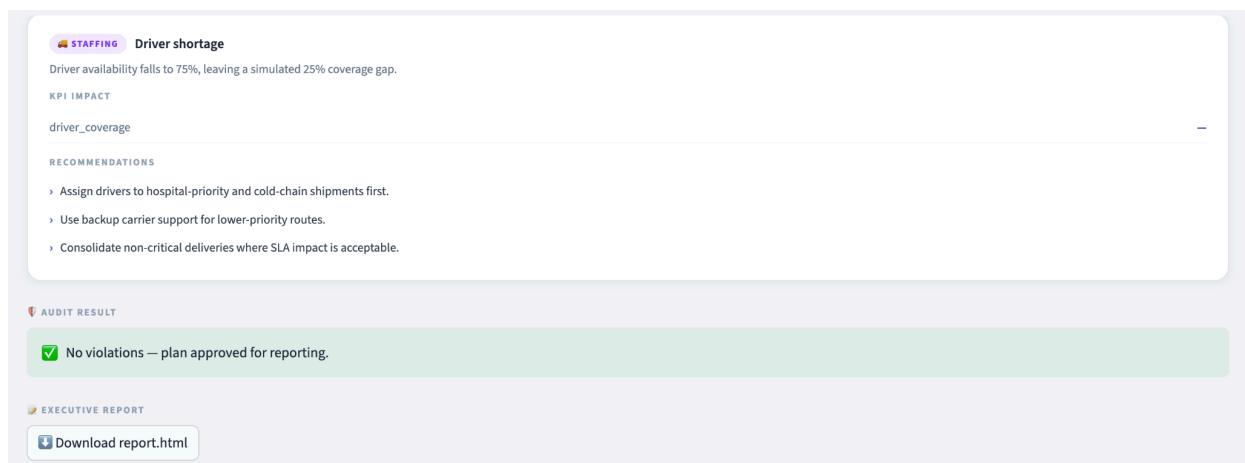


Figure B5d. Executive Report Generation and Final Operational Output

After successful audit validation, the system generates an executive-ready operational dispatch report summarizing weather risks, operational disruptions, shipment constraints, and recommended dispatch actions.

The final report is only produced after deterministic safety enforcement confirms policy compliance, thereby ensuring that unsafe or incomplete dispatch plans are blocked before executive delivery.



Dispatch Report: NJ to Boston Corridor

Reporting Period: Next 24 Hours

1) Executive Decision: **HOLD FOR REVIEW**

Full dispatch is currently on **HOLD** due to critical data quality issues affecting **48.3% of incoming shipments**. These issues prevent accurate tracking, ensure patient safety, and compromise dispatch planning. Resolution of these data issues is paramount before proceeding with dispatch.

2) Top Risks

- Critical Data Quality Impairment:** 42 out of 87 incoming shipment rows (48.3%) are excluded from dispatch due to fundamental data errors (missing unique IDs, duplicates, dosage ambiguity, unknown item IDs). This directly impacts dispatchability, patient safety, and operational efficiency.
- Moderate Weather Impact:** A route risk score of 1 is identified due to high wind gusts (51.5-52.2 km/h) at Newark (W1) and Bronx (W2). This necessitates a 10% travel time buffer, adding to transit times and increasing potential for further weather-related escalations.
- Operational Vulnerability:** Simulated scenarios reveal high-severity impacts from potential demand spikes (capacity utilization to 94%), primary warehouse closure (estimated 6-hour routing delay), and driver shortages (25% coverage gap). These indicate significant service disruption risks without robust contingency execution.

3) Recommended Actions

- **Immediate Data Remediation:**

- Resolve 5 missing unique_item_ids.
- Correct 10 duplicate unique_item_ids.
- Confirm dosage for 26 item_id 10021 records (patient safety risk).
- Register item_id 10071 in Item Master.

Owner: Data Operations Lead | **Timing:** Immediate (within 2 hours)

- **Proactive Contingency Planning Activation:**

- **Demand Spike:** Prioritize Tier 1 (Life-critical) shipments; reserve backup cold-chain capacity.
- **Warehouse Closure:** Formalize rerouting plans to alternate staging; escalate critical shipments for expedited transport.
- **Driver Shortage:** Prioritize driver assignments for hospital-critical and cold-chain shipments; engage backup carriers.

Owner: Logistics Manager / Operations Director | **Timing:** Pre-dispatch / Ongoing

- **Enhanced Data Ingestion Controls:** Implement automated validation checks to prevent future occurrences of missing/duplicate `unique_item_ids` and unknown `item_ids`.

Owner: IT/Data Engineering | **Timing:** Next 2 weeks

4) KPI Snapshot

KPI	Value	Threshold	Status
Excluded Row Count	42 rows (48.3%)	<5%	Critical
Logged Excluded Rows	42 rows	100% of excluded	Normal
Reported Excluded Rows	42 rows	100% of excluded	Normal
Weather Risk Summary	Route Risk Score: 1	0	Elevated
Applied Travel Buffers	10%	0% for Risk 0	Normal
Valid vs. Excluded Shipments	45 vs 42	Excluded <5%	Critical
SLA Risk Flags	To be assessed	0 flags	Normal (process in place)

Weather Details (I-95 Corridor)

Overall Route Risk Score: **1** (Elevated)

Waypoint	City, State	Risk Score (0-3)	Max Precip (mm/day)	Max Wind Gust (km/h)	Min Temp (°C)
W1	Newark, NJ	1	5.1	51.5	12.8
W2	Bronx, NY	1	3.6	52.2	12.5
W3	New Haven, CT	0	1.1	41.8	11.7
W4	Providence, RI	0	1.5	37.8	11.9
W5	Boston, MA	0	0.0	43.9	10.4

Note: Highlighted row indicates the worst waypoint (W1: Newark, NJ) contributing to the corridor's maximum risk score.

5) Audit Status: PASS

The dispatch plan successfully passed all audit checks against defined business rules, playbooks, and scenario constraints.

Corrections Made:

- The plan explicitly acknowledges and details the high volume of data quality issues identified by OpsDataAgent.
- Concrete actions are recommended to address each specific data quality issue.
- Contingency strategies for high-impact scenarios (demand spike, warehouse closure, driver shortage) from ScenarioAgent are integrated into the recommended actions.
- Weather buffer policy (10% for risk score 1) is correctly applied.

Caveats:

- Full dispatch remains contingent on the successful and timely resolution of the identified critical data quality issues.
- Continued monitoring of weather conditions is required, with escalation protocols in place for increased risk scores.

